Sommary sheet midterm 2

Interval	Set
$[a,b]$	$\{x \in R : a \le x \le b\}$
(a,b)	$\{x \in R : a < x < b\}$
$[a,b)$	$\{x \in R : a \le x < b\}$
$(a,b]$	$\{x \in R : a < x \le b\}$
$(a,+\infty]$	$\{x \in R : x > a\}$
$(-\infty,b)$	$\{x \in R : x < b\}$
$(-\infty,b]$	$\{x \in R : x \le b\}$
$(-\infty,+\infty)$	R

absolute values

$$|x| = \begin{cases} -x & \text{if } x<0 \\ x & \text{if } x \ge 0 \end{cases}$$

- $|x| \le t \iff -t \le x \le t$
- $|x| \ge t \iff x \ge t$ or $x \le -t$
- Triangle inequality: Let a,b be real numbers. Then $|a+b| \le |a| + |b|$

properties of exponents

- $x^0 = 1 \quad x \neq 0$
- $x^{-n} = \frac{1}{x^n} \quad x \neq 0$
- $x^{1/n} = \sqrt[n]{x}$ & $x^{m/n} = \sqrt[n]{x^m}$
- $x^m x^n = x^{m+n}$ & $x^m / x^n = x^{m-n}$
- $(x^m)^n = x^{mn}$
- $x^n y^n = (xy)^n$

Factoring polynomials

- $a^2 - b^2 = (a-b)(a+b)$
- $a^3 - b^3 = (a+b)(a^2 - ab + b^2)$
- $x^n - y^n = (x-y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \dots + xy^{n-2} + y^{n-1})$
- $(a \pm b)^2 = a^2 \pm 2ab + b^2$
- $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
- $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

rationalizing

- Denominator is $\sqrt{a}$, multiply top & bottom by $\sqrt{a}$
- Denominator is $\sqrt{a} \pm \sqrt{b}$, multiply top & bottom by $\sqrt{a} \mp \sqrt{b}$

average rate of change

Definition: rate of change of a function over an interval. Corresponds to the slope connecting the two endpoints of the interval.

Slope or Rate of change of y with respect to x through two points (x_1,y_1) and (x_2,y_2):

$$m = \frac{\text{the change in } y}{\text{the change in } x} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Instantaneous rate of change

Definition: rate of change at a specific point

* Slope of the tangent passing through a single point, or tangent point, on the graph of a function.

estimation: $\frac{f(a+h) - f(a)}{(a+h) - a}$

Limits of sequence

sequence: $a_1, a_2, \dots, a_n, \dots$

a_n is the nth term

Definition: If a sequence an arbitrarily closes to a unique number L as n goes to ∞, we say that a_n converges to L, which is written as: $\lim_{n\to\infty} a_n = L$

otherwise the sequence diverges

Limits of Functions

Function: relationship between two variables (d.v & i.v)

Domain → D Range → R

$\lim_{x\to a} f(x) = L$ → finite number

one-sided limit

- x approaches from left: $\lim_{x\to a^-} f(x) = L$
- x approaches from right: $\lim_{x\to a^+} f(x) = L$

continuity

Def: if $\lim_{x\to a} f(x) = f(a)$, then $f(x)$ is continuous at $x=a$, otherwise, $f(x)$ is discontinuous at $x=a$

- Jump discontinuity: one sided limits exist, but not equal
- Infinite discontinuity: left side or right side limit does not exist
- Removable discontinuity: one sided limits exist, and equal, but the function is undefined at $x=a$

First principles definition of the derivative

Definition: The derivative of the function $y = f(x)$ is the function $f'(x)$:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

meaning: $f'(a) =$

- instantaneous rate of change of $f(x)$ at a, or
- the slope of the tangent line to the curve at a

Some differentiation Rules

- Constant rule: if $f(x) = c$, then $f'(x) = 0$ or $\frac{d}{dx}(c) = 0$
- constant multiple rule: $[cf(x)]' = cf'(x)$
- Sum rule and difference rule: $[f(x) \pm g(x)]' = f'(x) \pm g'(x)$
- Power Rule: If $f(x) = x^n$, n is any real number. Then $f'(x) = nx^{n-1}$

The product rule

- Product rule:

$$[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$$

The quotient rule

- quotient rule:

$$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

The chain rule

- chain rule: $[f(g(x))]' = f'(g(x))g'(x)$, $\frac{df(g(x))}{dx} = \frac{df(u)}{du} \cdot \frac{dg(x)}{dx}$, $u = g(x)$, $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$
- General power rule: $[u(x)^n]' = nu^{n-1}u'(x)$

Velocity, acceleration, and second derivatives

Let $y = f(x)$. Then

$$y''(x) = f''(x) = \frac{d}{dx}\left(\frac{df}{dx}\right) = \frac{d}{dx}\left(\frac{dy}{dx}\right), \quad y^{(n)}(x) = f^{(n)}(x) = \frac{d}{dx}\left(\frac{dy^{(n-1)}}{dx}\right)$$

$$(x^n)^{(n)} = n!, \quad \left(\frac{1}{x}\right)^{(n)} = (-1)^n n! x^{-n-1}$$

Definition: Let $s = f(t)$ be the position function. s is called the displacement at time t.

velocity at t: $v(t) = f'(t)$

acceleration at t: $a(t) = f''(t) = v'(t)$

Increasing and decreasing functions

Increasing/decreasing test

- If $f'(x) > 0$ on an interval, then f is increasing on that interval
- If $f'(x) < 0$ on an interval, then f is decreasing on that interval
- If $f'(x) = 0$ on an interval, then f is a constant on that interval

Critical number

Definition: A point p in the domain such that $f'(p) = 0$ or $f'(p)$ undefined is called a critical number

$(p, f(p))$ → critical point $f(p)$ → critical value

Rational Functions

Def:

line $x=a$ is a vertical asymptote of the curve $y = f(x)$

$$\lim_{x\to a^-} f(x) = \pm\infty, \quad \lim_{x\to a^+} f(x) = \pm\infty, \quad \lim_{x\to a} f(x) = \pm\infty$$

Ex:

$$\lim_{x\to 0} \frac{1}{x^2} = \infty, \quad \lim_{x\to 1} \frac{-1}{(x-1)^2} = -\infty$$

Def:

If $\lim_{x\to\pm\infty} f(x) = L$, $y = L$ → H.A.

Ex:

$$\lim_{x\to\infty} \frac{1}{x^r} = 0, \ r > 0$$

Curve Sketching

1. Even function

- $f(-x) = f(x)$ for all x in D
- Curve is symmetric about the y-axis.

2. Odd function

- $f(-x) = -f(x)$ for all x in D
- Curve is symmetric about the origin

Sketch polynomial steps

1. Determine domain of function
2. Determine the intercepts of the function
3. Determine/classify critical numbers of the function, intervals of increase and decrease
4. Determine intervals of concavity & the points of inflection
5. Sketch the function

Rate of change problems

Functions pertaining to Business

x: number of units of a product or service sold

$p(x)$: price per unit

- Demand function: $p(x)$
- Revenue function: $R(x) = xp(x)$
- Cost function: total cost of producing x units of a product or service. $C(x)$
- Profit function: $P(x) = R(x) - C(x)$

Profit from the sale of x units of a product or service

Maxima and Minima

- Local (relative) extrema

local min: if $f(p) \le f(x)$ for points near p

local max: if $f(p) \ge f(x)$ for points near p

- Absolute maximum and minimum

Abs Max: abs max at p if $f(p) \ge f(x)$ for all x in the domain

Abs Min: abs min at p if $f(p) \le f(x)$ for all x in the domain

First derivative test

Let p → critical number

If f' changes from − to + (local min at p)

If f' changes from + to − (local max at p)

Fermat Theorem

If f has a relative max or min at c, and if $f'(c)$ exists, then $f'(c) = 0$

Closed Interval Method

To find a global max or min for $f(x)$ on a closed interval $[a,b]$:

1. Find all the critical numbers, e.g.: $x_1, \dots, x_n$
2. global minimum = $\min\{f(x_1), \dots, f(x_n), f(a), f(b)\}$, global maximum = $\max\{f(x_1), \dots, f(x_n), f(a), f(b)\}$.

Derivatives of Business functions

- $C'(x)$ → marginal cost function
- $R'(x)$ → marginal revenue function
- $P'(x)$ → marginal profit function

Concavity and the Second Derivative Test

Concavity test:

If $f''(x) > 0$ for all x in I, then the graph is concave upward

If $f''(x) < 0$ for all x in I, then the graph is concave downward

point of inflection

If $f(x)$ changes concavity at p, then p is an inflection point → $f''(p) = 0$ or undefined

Second derivative test:

- If $f''(p) > 0$, then f has a local minimum at p
- If $f''(p) < 0$, then f has a local maximum at p
- If $f''(p) = 0$, then nothing

Optimization problems

1. If possible, draw a sketch or diagram of the problem to be solved.
2. Define variables to be used & label the diagram with them
   * This helps to find/create mathematical equations
3. Write down all equations related to the problem or diagram.
   * make sure you know what to maximize/minimize
4. Express equations as a function in only one variable. Then differentiate the function.
5. Verify that your result is a maximum or minimum value using the first or second derivative test for extrema

Sinusoidal functions

Radian $\hookrightarrow$ Degree: t degree $= \frac{t}{180}\pi$

Consider a right triangle:

$$\sin t = \frac{\text{opposite}}{\text{hypothenuse}} \qquad \cos t = \frac{\text{adjacent}}{\text{hypothenuse}}$$

Basic relations:

$$\sin\left(\theta + \frac{\pi}{2}\right) = +\cos\theta \qquad \sin(\theta+\pi) = -\sin\theta$$
$$\cos\left(\theta + \frac{\pi}{2}\right) = -\sin\theta \qquad \cos(\theta+\pi) = -\cos\theta$$

Pythagorean trigonometric identity: $\sin^2 x + \cos^2 x = 1$

special values:

t	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin t$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
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Addition formulas:

$$\sin(x+y) = \sin x\cos y + \cos x \sin y$$
$$\cos(x+y) = \cos x\cos y - \sin x \sin y$$

Double-angle formulas:

$$\sin 2x = 2\sin x \cos x, \quad \cos 2x = \cos^2 x - \sin^2 x$$

Half-angle formula:

$$\sin^2 x = \frac{1-\cos 2x}{2}, \quad \cos^2 x = \frac{1+\cos 2x}{2}$$

Periods: $\sin x$ and $\cos x$ have period 2π, $\tan x$ and $\cot x$ have period π

Special inequalities:

$$-|x| \le \sin x \le |x|, \quad -|x| \le 1-\cos x \le |x|$$

Derivatives of the sine & cosine functions

$$\lim_{h\to 0} \frac{\sinh}{h} = 1$$

derivative of sine and cosine functions:

$$(\sin x)' = \cos x, \quad (\cos x)' = -\sin x$$

Final exam MAT 1339

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absolute values

$$|x| = \begin{cases} -x & \text{if } x<0 \\ x & \text{if } x \ge 0 \end{cases}$$

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- $|x| \ge t \iff x \ge t$ or $x \le -t$
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Function: relationship between two variables (d.v & i.v)
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Some differentiation Rules

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- Chain rule: $[f(g(x))]' = f'(g(x))g'(x)$, $\frac{df(g(x))}{dx} = \frac{df(x)}{dv} \cdot \frac{dg(x)}{dx}$, $v = g(x)$, $\frac{dy}{dx} = \frac{dy}{dv} \cdot \frac{dv}{dx}$
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Velocity, acceleration, and second derivatives

Let $y = f(x)$. Then
$y''(x) = f''(x) = \frac{d}{dx}\left(\frac{df}{dx}\right) = \frac{d}{dx}\left(\frac{dy}{dx}\right)$, $y^{(n)}(x) = f^{(n)}(x) = \frac{d}{dx}\left(\frac{dy^{(n-1)}}{dx}\right)$.

$(x^n)^{(n)} = n!$, $\left(\frac{1}{x}\right)^{(n)} = (-1)^n n! x^{-n-1}$

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velocity at t: $v(t) = f'(t)$

acceleration at t: $a(t) = f''(t) = v'(t)$

Increasing and decreasing functions

Increasing/decreasing test
- If $f'(x) > 0$ on an interval, then f is increasing on that interval
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- If $f'(x) = 0$ on an interval, then f is a constant on that interval

Critical number
Definition: A point p in the domain such that $f'(p) = 0$ or $f'(p)$ undefined is called a critical number
$(p, f(p))$ → critical point $\quad f(p)$ → critical value

Rate of change problems

Functions pertaining to Business

x: number of units of a product or service sold
$p(x)$: price per unit

- Demand function: $p(x)$
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  Profit from the sale of x units of a product or service

Derivatives of Business functions

- $C'(x)$ → marginal cost function
- $R'(x)$ → marginal revenue function
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Maxima and Minima

- Local (relative) extrema
  - local min: if $f(p) \le f(x)$ for points near p
  - local max: if $f(p) \ge f(x)$ for points near p
- Absolute maximum and minimum
  Abs Max: abs max at p if $f(p) \ge f(x)$ for all x in the domain
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First derivative test
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Fermat Theorem
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Concavity and the Second Derivative Test

Concavity test:
If $f''(x) > 0$ for all x in I, then the graph is concave upward
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point of inflection
If $f(x)$ changes concavity at p, then p is an inflection point
→ $f''(p) = 0$ or undefined

Second derivative test:
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Rational Functions

Def:
line $x = a$ is a vertical asymptote of the curve $y = f(x)$

$$\lim_{x \to a^-} f(x) = \pm\infty, \quad \lim_{x \to a^+} f(x) = \pm\infty, \quad \lim_{x \to a} f(x) = \pm\infty$$

Ex:

$$\lim_{x \to 0} \frac{1}{x^2} = \infty, \quad \lim_{x \to 1} \frac{-1}{(x-1)^2} = -\infty$$

Def:
If $\lim_{x \to \pm\infty} f(x) = L$, $y = L$ → HA.

Ex:

$$\lim_{x \to \infty} \frac{1}{x^r} = 0, \ r > 0$$

Curve Sketching

1. Even function

- $f(-x) = f(x)$ for all x in D
- Curve is symmetric about the y-axis.

2. Odd function

- $f(-x) = -f(x)$ for all x in D
- Curve is symmetric about the origin

Sketch polynomial steps

1. Determine domain of function
2. Determine the intercepts of the function
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Sinusoidal functions

Radian ⇆ Degree: $+$ degree $= \frac{+}{180}\pi$

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Addition formulas: $\sin(x+y) = \sin x\cos y + \cos x \sin y$

$\cos(x+y) = \cos x \cos y - \sin x \sin y$

Double-angle formulas:

$\sin 2x = 2\sin x \cos x$, $\cos 2x = \cos^2 x - \sin^2 x$

Half-angle formula:

$\sin^2 x = \frac{1-\cos 2x}{2}$, $\cos^2 x = \frac{1+\cos 2x}{2}$

Periods: $\sin x$ and $\cos x$ have period 2π, $\tan x$ and $\cot x$ have period π

Special inequalities:

$-|x| \le \sin x \le |x|$, $\quad -|x| \le 1-\cos x \le |x|$

Derivatives of the sine & cosine functions

$$\lim_{h\to 0} \frac{\sinh}{h} = 1$$

derivative of sine and cosine functions:

$(\sin x)' = \cos x$, $\quad (\cos x)' = -\sin x$

Exponential & logarithmic functions

$y = f(x) = a^x$

- Expo. growth: $a > 1$: $\lim_{x\to-\infty} a^x = 0$, $\lim_{x\to\infty} a^x = \infty$
- Expo. decay: $0 < a < 1$: $\lim_{x\to-\infty} a^x = \infty$, $\lim_{x\to\infty} a^x = 0$

Laws:

$a^{x+y} = a^x a^y$, $a^{x-y} = a^x/a^y$, $(a^x)^y = a^{xy}$, $a^x b^x = (ab)^x$

Compounded interest:

- Compounded n times a year: $A = P(1+\frac{r}{n})^{nt}$
- Compounded continuously: $A = Pe^{rt}$

$y = a^x \to y = \log_a x$, $y = e^x \to y = \log_e x = \ln x$, $y = 10^x \to y = \log_{10} x = \log x$

laws:

$\ln(BC) = \ln B + \ln C$

$\ln(\frac{B}{C}) = \ln B - \ln C$

$\ln(B^n) = n\ln B$

$\ln(e^x) = x$, $\ln e = 1$

$e^{\ln B} = B$

$\log_e 1 = 0$

Derivatives of exponential & logarithmic functions

$(e^x)' = e^x$, $(e^{u(x)})' = e^{u(x)}u'x$

Derivatives of log functions

Derivatives of log functions

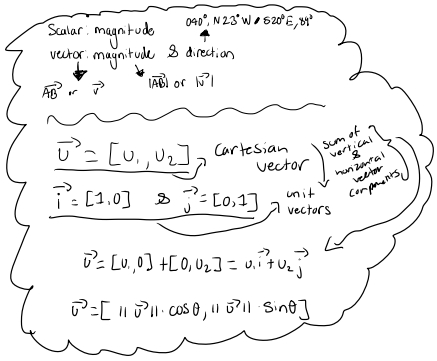
$\frac{d}{dx}(\ln x) = \frac{1}{x}$, $(\ln f(x))' = \frac{f'(x)}{f(x)}$

$(\log_a |x|)' = \frac{1}{x\ln a}$, $(\log_a f(x))' = \frac{f'(x)}{f(x)\ln a}$

$(a^x)' = a^x(\ln a)$, $[a^{v(x)}]' = a^{v(x)}(\ln a)\,v'(x)$

change base

$\log_a b = \frac{\ln b}{\ln a}$



Dot product: $\vec{u} = [u_1, u_2]$, $\vec{v} = [v_1, v_2]$

$\vec{u}\cdot\vec{v} = u_1v_1 + u_2v_2$

Angle: $* \quad 0 \le \theta \le \pi$

$\cos\theta = \frac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|\|\vec{v}\|}$

Orthogonal: $\vec{u} \perp \vec{v}$ if $\vec{u}\cdot\vec{v} = 0$

Properties:

- $\vec{u}\cdot\vec{v} = \vec{v}\cdot\vec{u}$
- $\vec{w}\cdot(\vec{u}+\vec{v}) = \vec{w}\cdot\vec{u} + \vec{w}\cdot\vec{v}$
- $c(\vec{u}\cdot\vec{v}) = (c\vec{u})\cdot\vec{v} = \vec{u}\cdot(c\vec{v})$
- $\vec{0}\cdot\vec{u} = 0$
- $\vec{u}\cdot\vec{u} = \|\vec{u}\|^2$

magnitude: $\|[a,b]\| = \sqrt{a^2+b^2}$. Unit vector: $|\vec{u}| = 1$

Addition: $\vec{u} = [u_1, u_2]$, $\vec{v} = [v_1, v_2]$

$\vec{u} + \vec{v} = [u_1 + v_1, u_2 + v_2]$

Subtraction: $\vec{u} = [u_1, u_2]$, $\vec{v} = [v_1, v_2]$

$\vec{u} - \vec{v} = [u_1 - v_1, u_2 - v_2]$

Scalar multiplication: c, $\vec{u} = [u_1, u_2]$, $\vec{v} = [v_1, v_2]$

$c\vec{u} = [cu_1, cu_2]$

vector projections: projection of $\vec{v}$ on $\vec{u}$

$\text{Proj}_{\vec{u}}\vec{v} = \left(\frac{\vec{v}\cdot\vec{u}}{\vec{u}\cdot\vec{u}}\right)\vec{u}$

Equation of lines: point & nonzero

- Equation in standard (scalar equation): $Ax + By + C = 0$
- Vector equation: $[x, y] = [x_0, y_0] + t[v_1, v_2]$, $t \in \mathbb{R}$
- Parametric equation: $x = x_0 + tv_1$, $y = y_0 + tv_2$

$\vec{v} = [v_1, v_2, v_3]$

$\vec{i} = [1,0,0]$, $\vec{j} = [0,1,0]$, $\vec{k} = [0,0,1]$

$\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$

Magnitude: $|[a,b,c]| = \sqrt{a^2+b^2+c^2}$

↳ unit vector: $|\vec{v}| = 1$

Addition: $\vec{u} = [u_1, u_2, u_3]$, $\vec{v} = [v_1, v_2, v_3]$

$\vec{u} + \vec{v} = [u_1 + v_1, u_2 + v_2, u_3 + v_3]$

Subtraction: $\vec{u} = [u_1, u_2, u_3]$, $\vec{v} = [v_1, v_2, v_3]$

$\vec{u} - \vec{v} = [u_1 - v_1, u_2 - v_2, u_3 - v_3]$

Scalar multiplication: $\vec{u} = [u_1, u_2, u_3]$, c

$c\vec{u} = [cu_1, cu_2, cu_3]$

Dot product: $\vec{u} = [u_1, u_2, u_3]$, $\vec{v} = [v_1, v_2, v_3]$

$\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2 + u_3v_3$

Angle: $0 \leq \theta \leq \pi$, $\cos\theta = \dfrac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|\cdot\|\vec{v}\|}$

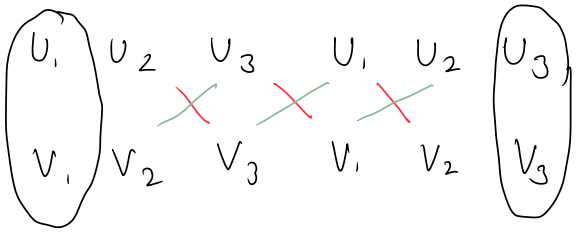
θ → angle between $\vec{u}$ & $\vec{v}$

projection of $\vec{v}$ on $\vec{u}$:

$$\text{Proj}_{\vec{u}}\vec{v} = \left(\frac{\vec{v}\cdot\vec{u}}{\vec{u}\cdot\vec{u}}\right)\vec{u}$$

cross product: $\vec{u} = [u_1, u_2, u_3]$, $\vec{v} = [v_1, v_2, v_3]$

$\vec{u} \times \vec{v} = [u_2v_3 - u_3v_2, +u_3v_1 - u_1v_3, u_1v_2 - u_2v_1]$



rule: for each cross (x), down minus up

* area of parallelogram by $\vec{u}$ & $\vec{v}$: $|\vec{u} \times \vec{v}|$
* volume of parallelepiped by $\vec{u}$, $\vec{v}$ & $\vec{w}$: $|\vec{u} \cdot (\vec{v} \times \vec{w})|$

Equation of lines

$P(x_0, y_0, z_0)$ ← point

$\vec{v} = [v_1, v_2, v_3]$ ← nonzero direction vector

- Vector equation: $[x,y,z] = [x_0, y_0, z_0] + t[v_1, v_2, v_3]$, $t \in \mathbb{R}$
- Parametric equation: $x = x_0 + tv_1$, $y = y_0 + tv_2$, $z = z_0 + tv_3$

Planes

Def: a plane Π is determined by a point, $P(x_0, y_0, z_0)$, and two non-parallel direction vectors parallel to the plane (or a normal vector $\vec{n}$ which is perpendicular to the plane).

- Vector equation: $\vec{r} = \vec{r_0} + t\vec{a} + s\vec{b}$
  - $\vec{r} = [x,y,z]$ is a position vector for any point (x,y,z) on the plane
  - $\vec{r_0} = [x_0, y_0, z_0]$ is the position vector for the given point (x_0, y_0, z_0) on the plane
  - $\vec{a}$, $\vec{b}$ are two non-parallel direction vectors parallel to the plane
  - s, t are scalars in $\mathbb{R}$
- Parametric equations:

  $x = x_0 + ta_1 + sb_1$, $y = y_0 + ta_2 + sb_2$, $z = z_0 + ta_3 + sb_3$

  $\vec{a} = [a_1, a_2, a_3]$, $b = [b_1, b_2, b_3]$
- Scalar equation (standard equation):

  $Ax + By + Cz + D = 0$

  $\vec{n} = [A, B, C] = \vec{a} \times \vec{b}$

Intersections of lines

L_1 & L_2	$\mathbb{R}^2$	$\mathbb{R}^3$
	• parallel • intersected	• parallel • intersected • skewed

Distance:

- distance from a point B to a line L: $\vec{r} = \vec{p} + t\vec{a}$

  $d(B, L) = |\vec{BP} - \text{proj}_{\vec{a}} \vec{BP}|$

- distance between two parallel lines: $L_1: \vec{r} = \vec{p} + t\vec{a}$ & $L_2: \vec{r} = \vec{q} + t\vec{a}$

  $$d(L_2, L_2) = d(B, L_1)$$

- distance between two skew lines: $L_1: \vec{r} = \vec{p} + t\vec{a}$, $L_2: \vec{r} = \vec{q} + t\vec{b}$

  $\vec{n} = \vec{a} \times \vec{b}$ (normal vector perpendicular to L_1 & L_2)

  $$d = |\text{proj}_{\vec{n}} \vec{pq}|$$

Intersections of lines & planes

Intersections between a line & plane:

$\Pi: Ax + By + Cz + D = 0$, $\vec{n} = [A, B, C]$ (normal vector to the plane)

$L: [x, y, z] = [x_0, y_0, z_0] + t\vec{v}$, $t \in \mathbb{R}$

- If $\vec{n} \cdot \vec{v} = 0$, then $\Pi \parallel L$. (either the line is on the plane or no intersection)
- If $\vec{n} \cdot \vec{v} \neq 0$, only one intersection

Distance between a point and a plane:

plane Π is determined by a point and a normal vector $\vec{n}$ which is perpendicular to the plane.

$P(p_1, p_2, p_3)$, $\Pi: ax + by + cz + d = 0$.

$$D = \frac{|ap_1 + bp_2 + cp_3 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

Intersections between two planes:

Possibilities

- intersect in a line (infinite number of solutions)
- planes are coincident (infinite solutions)
- planes are parallel and distinct (no solution)

Intersections between three planes:

$\vec{n}_1, \vec{n}_2, \vec{n}_3$ are coplanar if $\vec{n}_1 \cdot (\vec{n}_2 \times \vec{n}_3) = 0$

Possibilities

- Planes intersect in a line
  $\vec{n}_1 \cdot (\vec{n}_2 \times \vec{n}_3) = 0$
  normals are coplanar
  and the planes intersect in a line or not at all
- Planes intersect in a point.
  $\vec{n}_1 \cdot (\vec{n}_2 \times \vec{n}_3) \neq 0$
  normals are not coplanar
- There is no solution
  $\vec{n}_1 \cdot (\vec{n}_2 \times \vec{n}_3) = 0$
  Planes intersect in a line or not at all